

- 6 The following table shows part of the instruction set for a processor. The processor has two registers: the Accumulator (ACC) and the Index Register (IX).

Instruction		Explanation
Opcode	Operand	
LDM	#n	Immediate addressing. Load the number n to the ACC
AND	#n / Bn / &n	Bitwise AND operation of the contents of the ACC with the operand
AND	<address>	Bitwise AND operation of the contents of the ACC with the contents of <address>
XOR	#n / Bn / &n	Bitwise XOR operation of the contents of the ACC with the operand
XOR	<address>	Bitwise XOR operation of the contents of the ACC with the contents of <address>
OR	#n / Bn / &n	Bitwise OR operation of the contents of the ACC with the operand
OR	<address>	Bitwise OR operation of the contents of the ACC with the contents of <address>
CMP	#n	Compare the contents of the ACC with number n
CMP	<address>	Compare the contents of the ACC with the contents of <address>
LSL	#n	Bits in the ACC are shifted logically n places to the left. Zeros are introduced on the right-hand end
LSR	#n	Bits in the ACC are shifted logically n places to the right. Zeros are introduced on the left-hand end
<address> can be an absolute or symbolic address # denotes a denary number, e.g. #127 B denotes a binary number, e.g. B10010001 & denotes a hexadecimal number, e.g. &4A		

The current contents of main memory are shown:

Address	Data
100	0000 0011
101	1010 1110
102	1100 1100
103	1111 1111
104	1100 1100

